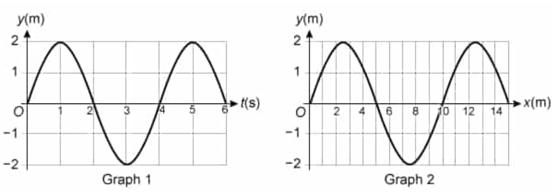
One end of a horizontal string is moved vertically and produces a transverse periodic wave that travels along the string in the *x*-direction. Graph 1 shows the displacement *y* of the moving end of the string over time. Graph 2 shows the relationship between wave position and distance *x* along the string at a particular time.

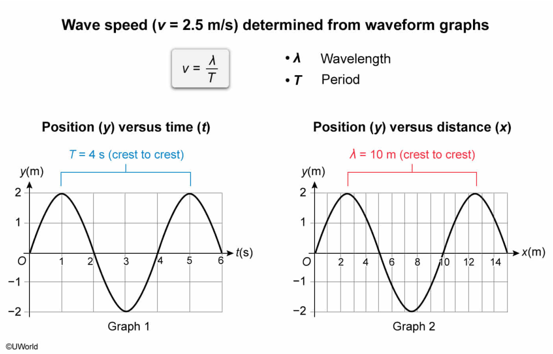


What is the speed of the wave?

- A. 0.4 m/s [5%]
- B. 2.5 m/s [79%]
- C. 20 m/s [8%]
- D. 40 m/s [6%]

Correct
79% answered correctly

Explanation:



A perturbation in a medium can produce a traveling wave. As the wave passes through a given point in the medium, the wave displaces the medium at that point. For transverse waves, displacements in the medium move perpendicular to the direction of wave propagation. The motion of each displacement is oscillatory, and the wave period (T) is the **time** it takes to complete one cycle (eg, move down from one crest and then back up to the next crest).

The wave also displays periodic behavior in space. The wavelength (λ) is the **distance** between successive crests or troughs, or (more generally) the distance between two neighboring points in the medium with the same behavior at a given time.

The **speed of the wave** v as it propagates through the medium is the ratio of λ and T:

$$v = \frac{\lambda}{T}$$

In this question, Graph 1 shows the vertical position of a string element as a function of time during transverse wave motion. The time for the element to complete one cycle is T = 4 s.

Graph 2 shows a snapshot of the string at a given time and the distance from crest to crest is λ = 10 m. Therefore, the speed of the wave is:

$$v = \frac{\lambda}{T} = \frac{10\ \mathrm{m}}{4\ \mathrm{s}} = 2.5\ \mathrm{m/s}$$

(Choice A) Dividing T by λ produces a value of 0.4 with units of s/m. Speed has units of m/s.

(Choice C) This answer is the product of wave amplitude and wavelength. Amplitude has no effect on wave speed.

(Choice D) Multiplying λ by T yields a value of 40 with units of m·s. The correct units for speed are m/s.

Educational objective:
The wavelength is the distance over which a wave repeats, and the period is the time in which a wave repeats. Wave speed can be determined by dividing the wavelength by the period.

Time spent:0 sec	**QID**:403906
Subject: Physics	**System:** Light and Sound